THE DIVISOR OF A NONZERO GLOBAL SECTION EQUALS THE CLASS OF ITS ZERO SCHEME AND COMPUTES THE FIRST CHERN CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an elementary etale neighbourhood trivialising the sheaf, the identification of the zero scheme with the zero scheme of a local equation, and a length comparison through the henselization

Editorial note. The label lemma-compute-c1 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The zero scheme of a nonzero global section of an invertible sheaf and the divisor of a section, together with the order of vanishing of a section along a prime divisor
- Elementary etale neighbourhoods and the identification of the henselizations of the local rings of a point and of the corresponding point of the neighbourhood
- Flat base change of lengths along the map from a local ring to its henselization, and the identification of the coefficients of a cycle by comparing them at generic points

Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Prerequisite pointer invoked at proof line(s) 2375; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0EMV: sources/stacks-chow/tags.txt, source lines 18320–18320. Prerequisite pointer invoked at proof line(s) 2350; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 2329; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0845: sources/stacks-chow/tags.txt, source lines 10310–10310. Prerequisite pointer invoked at proof line(s) 2357; it names a

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standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0EPR: sources/stacks-chow/tags.txt, source lines 18386–18386. Prerequisite pointer invoked at proof line(s) 2378; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0ENV: sources/stacks-chow/tags.txt, source lines 18355–18355. Prerequisite pointer invoked at proof line(s) 2358; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Assume X is integral and $n = \dim_\delta(X)$. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $s \in \Gamma(X, \mathcal{L})$ be a nonzero global section. Then*

$$div_{\mathcal{L}}(s) = [Z(s)]_{n-1}$$

in $Z_{n-1}(X)$ and

$$c_1(\mathcal{L}) \cap [X] = [Z(s)]_{n-1}$$

in $\mathrm{CH}_{n-1}(X)$.

Proof. Let $Z \subset X$ be an integral closed subspace of δ-dimension $n-1$. Let $\xi \in |Z|$ be its generic point. To prove the first equality we compare the coefficients of Z on both sides. Choose an elementary étale neighbourhood $(U, u) \to (X, \xi)$, see Decent Spaces, Section 0EMV and recall that $\mathcal{O}^h_{X,\xi} = \mathcal{O}^h_{U,u}$ in this case. After replacing U by an open neighbourhood of u we may assume there is a trivializing section s_U of $\mathcal{L}|_U$. Write $s|_U = f s_U$ for some $f \in \Gamma(U, \mathcal{O}_U)$. Then $Z \times_X U$ is equal to $V(f)$ as a closed subscheme of U, see Divisors on Spaces, Definition 0845. As in Spaces over Fields, Section 0ENV denote $\mathcal{L}_\xi$ the pullback of $\mathcal{L}$ under the canonical morphism $c_\xi : \mathrm{Spec}(\mathcal{O}^h_{X,\xi}) \to X$. Denote s_ξ the pullback of s_U; it is a trivialization of $\mathcal{L}_\xi$. Then we see that $c_\xi^*(s) = f s_\xi$. The coefficient of Z in $[Z(s)]_{n-1}$ is by definition

$$\mathrm{length}_{\mathcal{O}_{U,u}}(\mathcal{O}_{U,u}/f\mathcal{O}_{U,u})$$

Since $\mathcal{O}_{U,u} \to \mathcal{O}^h_{X,\xi}$ is flat and identifies residue fields this is equal to

$$\mathrm{length}_{\mathcal{O}^h_{X,\xi}}(\mathcal{O}^h_{X,\xi}/f\mathcal{O}^h_{X,\xi})$$

by Algebra, Lemma 02M1. This final quantity is equal to $\mathrm{ord}_{Z,\mathcal{L}}(s)$ by Spaces over Fields, Definition 0EPR, i.e., to the coefficient of Z in $\mathrm{div}_{\mathcal{L}}(s)$ as desired. □